

MATHEMATICS FOUNDATION-2

WEEK-2

$|AB| = |A| \cdot |B|$
 $|kA| = k^n |A| \rightarrow$ multiplying row by scalar $\times$ multiplies the determinant by k .

$\text{adj}(AB) = (\text{adj } B)(\text{adj } A)$
 $\text{adj}(kA) = k^{n-1} (\text{adj } A)$
 $|\text{adj } A| = |A|^{n-1}$
 $|\text{adj}(\text{adj } A)| = |A|^{(n-1)^2}$
 $\text{adj}(\text{adj } A) = |A|^{n-2} A$
 $\text{adj}(A^{-1}) = (\text{adj } A)^{-1} = \frac{A}{|A|}$

Properties of Δ :

- adding $R_1 \rightarrow R_1 + 2R_2$ does not change $\det$.
 $\rightarrow$ multiple of a row
- Switching 2 rows/cols changes the sign of $\det$.

(Inverse)

$$A^{-1} = \frac{\text{adj } A}{|A|}$$

- $|A| \neq 0 \rightarrow$ inverse exist (Non-singular)
- $|A| = 0 \rightarrow$ not exist (Singular)
- Gen. sys. $\rightarrow AX = B$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Homogenous sys: $AX = 0$

Prop: $\rightarrow$

- $(A^k)^{-1} = (A^{-1})^k = A^{-k}$
- $(A^T)^{-1} = (A^{-1})^T$
- $|A^{-1}| = \frac{1}{|A|}$

★

- $(AB)^{-1} = B^{-1}A^{-1}$
- $(kA)^{-1} = \frac{1}{k} \cdot A^{-1}$

Echelon Form

RREF $\rightarrow$ Leading non-zero entry is 1, and is the only non-zero entry in its column

Dependent var. $\rightarrow$ col. with a leading entry

Independent var. $\rightarrow$ col. without a leading entry

$$\text{no. of free variable} = \text{no. of var.} - \text{Rank}$$

$\hookrightarrow$ if present

$\hookrightarrow \infty$ many solutions

$\hookrightarrow$ no. of cols with pivots present

Q. Let $Ax=b$ be a matrix rep. of system of L.E, where A is a 4×4 matrix, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$, and $b = \begin{bmatrix} b_1 \\ b_2 \\ 0 \\ 0 \end{bmatrix}$

Let A be $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Which is correct option?

$x_1 \rightarrow$ D.V

$x_2 \rightarrow$ I.V

$x_3 \rightarrow$ D.V

$x_4 \rightarrow$ I.V

sol. I.V of Free var. is present
so, ∞ solutions

Transpose Prop.

1) $(A^T)^T = A$

2) $(A+B)^T = A^T + B^T$

3) $(kA)^T = k(A^T)$

★ 4) $(AB)^T = B^T A^T$

5) $(ABC)^T = C^T B^T A^T$

6) $|A^T| = |A|$